Marwan Ghazal

AI Student

Sheikh Zayed +20 101 651 1889 marwanghazal002y@gmail.com LinkedIn

Github

PROFILE

AI Engineering student at Nile University (GPA 4.0/4.0) building production AI systems with deep learning, LLMs, and RAG. Experience shipping end-to-end: a clinical neurofeedback platform built solo with an industry partner, two deployed automation systems from my NTA internship, and a 3rd-place Arabic NLP pipeline at DeepX 2026. Comfortable across the stack, from data pipelines and model training through FastAPI services and structured-output integrations with OpenAI, Gemini, and Claude APIs.

SKILLS

Languages
Python, JavaScript (Node.js), SQL

RAG & Retrieval
chunking/summarization, embeddings concepts, vector search, grounding & hallucination reduction

Databases & Platforms
PostgreSQL, MySQL, Supabase, Vercel

Dev Tools
Git/GitHub, Cursor, Copilot, ChatGPT, Windsurf

Backend & APIs
FastAPI, Flask, REST, JSON

LLMs / AI
OpenAI API, Gemini API, locally hosted Llama-family models via Hugging Face (Transformers), prompt engineering (system prompts, schema-based JSON outputs), cost-aware prompting.

Data/ML
Pandas, NumPy, scikit-learn, TensorFlow/Keras, XGBoost, Pytorch

PROFESSIONAL EXPERIENCE

AI Intern — National Training Academy (NTA) 08/2025 – 09/2025

- Delivered two production-ready AI systems: a **CV ranking pipeline** and an **OCR-based National ID extraction web app** for digital transformation workflows.
- Built an AI-powered CV ranking system using **Llama 3.1 + LoRA**, evaluating candidate–job fit via NLP analysis.
- Developed OCR pipelines using **OpenCV + Tesseract**, including preprocessing, correction, and extraction logic.

Impact: automated candidate shortlisting workflows and enabled reliable Arabic ID extraction via web APIs.

Junior Teaching Assistant — Nile University 02/2025 – 06/2025

- supported 50+ students with debugging and problem-solving guidance.
- Explained complex programming concepts clearly and efficiently, helping students resolve issues faster

PROJECTS

NeuroVista Pro — Clinical Neurofeedback Analysis Platform
Python, FastAPI, PostgreSQL + pgvector, LlamaIndex, Tauri, React

- Building a **desktop clinical platform** in partnership with **BioLife** (neurofeedback equipment provider) for analyzing EEG recordings, tracking patient progress, and recommending evidence-based protocols.
- Designed a layered architecture separating clinical logic from infrastructure: deterministic **decision engine**, multi-format **EEG parser** (EDF, proprietary CSV), and async FastAPI services backed by **PostgreSQL** with pgvector and tsvector.
- Implemented an **agentic RAG** clinical assistant with **hybrid retrieval** (semantic + keyword), cross-encoder reranking.

Arabic ABSA — DeepX 2026 Hackathon (3rd Place)
Python, PyTorch, MARBERTv2, Transformers, FastAPI, React
- Built a **25-head** joint multi-label Aspect-Based Sentiment Analysis system for Arabic customer reviews in a **12-hour** sprint. Co-developed with a 3-person team.
- Achieved **0.78 F1** on the hidden test set with a **custom training recipe** (Asymmetric Loss, layer-wise LR decay, EMA shadow weights, per-class threshold calibration) for severe class imbalance.
- Implemented intelligent language-aware routing: automatic script and language detection routes **Arabic** input to **MARBERTv2** and Latin-script input to **XLM-R** for multilingual coverage across **15+ languages**, with a rule-based fallback for noisy or ambiguous cases.
- Built multi-target translation **augmentation** (NLLB-200, opus-mt) expanding the training pool 6.9×, plus a unified preprocessing contract guaranteeing zero training-serving skew.

CV Ranking Automation (LLM + LoRA Adapter)
Python, Flask
- Built a CV-to-job ranking service using **Llama 3.1 + LoRA**, generating structured JSON scores for automation.
- Designed an **NLP pipeline** to analyze and rank CVs against job descriptions.
- Added **multilingual** support

Smart Food Guardian — Multimodal Nutri-Score (1st Place — Nile University UGRF)
Python, TensorFlow, XGBoost
- Developed a **multimodal machine learning system** combining tabular nutrition data, textual metadata, and product images.
- Demonstrated strong understanding of embeddings, modality fusion, and ML experimentation.
- Achieved 0.87 test accuracy / 0.83 macro-F1 on 5-class prediction.

Career Guidance AI System
Python, Flask, LLM APIs
- Built a quiz-based system that calls **Gemini API** for career assessment and reasoning.
- Implemented CV analysis workflows using **ChatPDF API** with structured AI responses.
- Designed REST endpoints for submission, retrieval, and persistence; stored user profiles, answers, and recommendation history in **PostgreSQL**.

Egyptian National ID OCR
FastAPI, OpenCV, Tesseract
- Built a hybrid OCR pipeline with preprocessing (grayscale/blur/adaptive thresholding/perspective correction) and field-level validation.
- Delivered a simple web interface + API for real-time extraction.

EDUCATION

B.Sc. in Artificial Intelligence — Nile University
Junior, Expected 2027
GPA: 4.0 | President's Honor (Spring 2025, Fall 2025)

EXTRACURRICULAR ACTIVITIES

Vice Head of Public Relations — Microsoft Student Club, Nile University (Current)
- Lead the PR team across the semester, owning team coordination, content direction, and execution of club initiatives.
- Part of the organizing team for DeepX 2026, the club's flagship hackathon, responsible for end-to-end speaker logistics and communications.

Head of Public Relations — E-Club, Nile University (Previous)
- Led PR strategy and execution for flagship events; managed sponsor and guest-speaker outreach.
- Trained PR team members on outreach messaging, communication etiquette, and content standards.